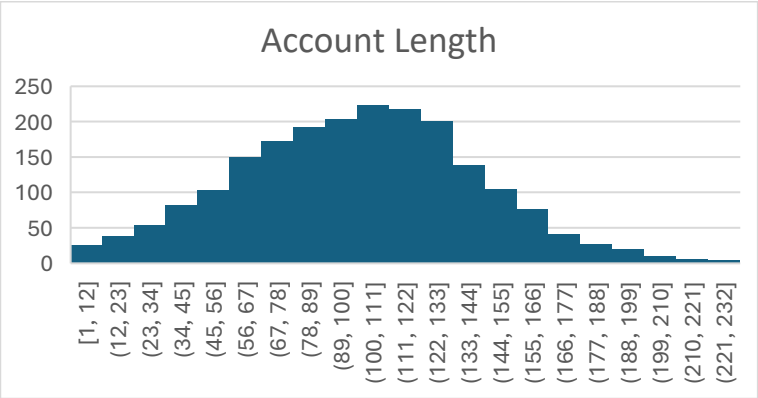
Business Analytics Coursework Report

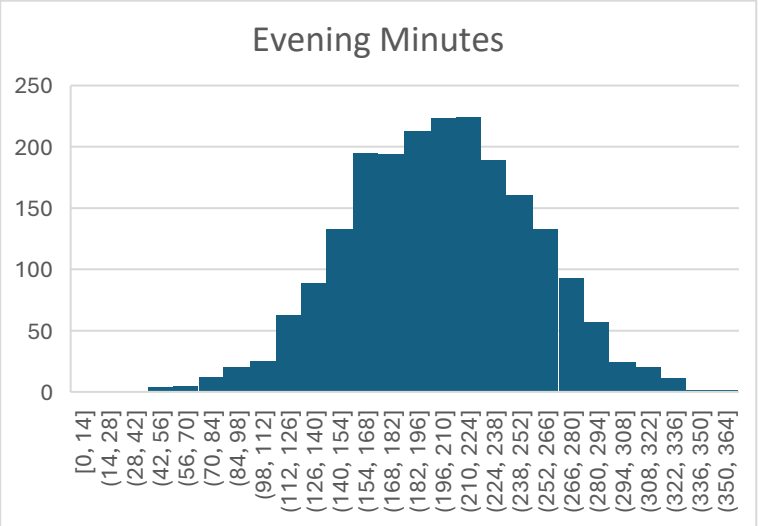
Problem 1: Customer Churn Analysis

In this part of the coursework, I have found the key factors that influence customer churn using a dataset from a telecom company. Initially, it was crucial to understand the distribution of variables, examine their relationships, assess their impact on churn, and provide business recommendations.

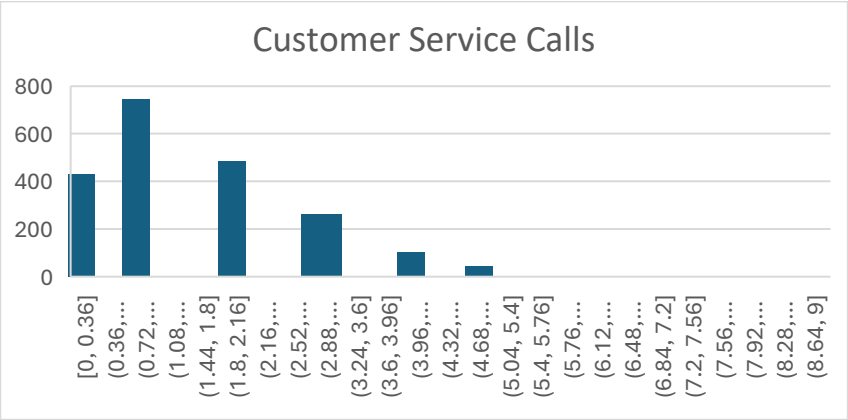
Part (a): Variable Distribution Analysis

The first step was to get the distribution of each numeric variable and create histograms for all columns from Account_Length to Total_Intl_Charge. Some variables, such as Account_Length and Total_Eve_Minutes, showed approximately symmetric distributions, indicating a normal curve.

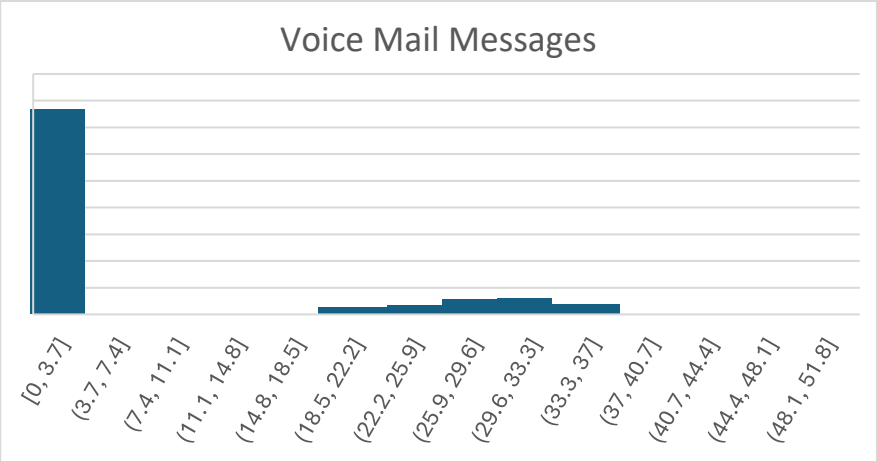


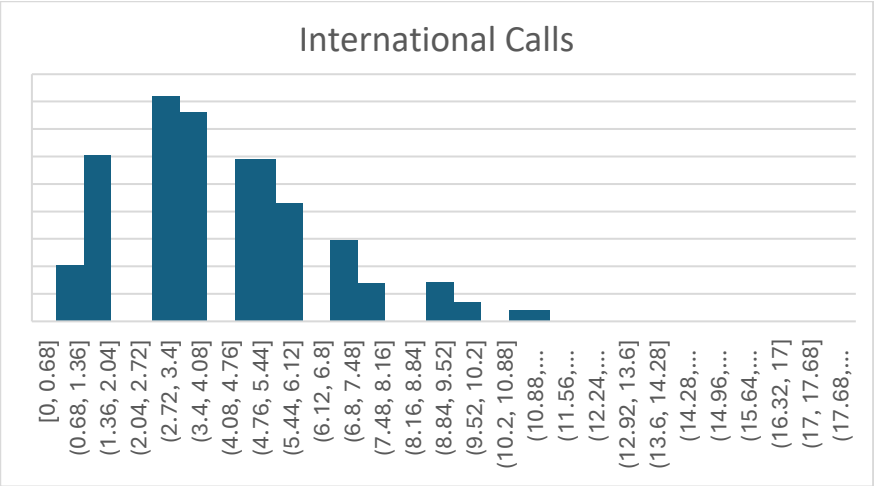


Others, like Customer_Service_Calls and all charge-based variables, were right-skewed.



Many variables had significant increases at zero—especially Number_Vmail_Messages, Customer_Service_Calls, and Total_Intl_Calls—which showed that a huge number of customers did not use voicemail or international services. This step helped me identify potential variables that could be related to customer dissatisfaction or disconnection.





Part (b): Inter-variable Correlation Analysis

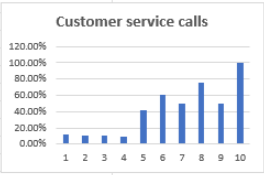
The next step was to find relationships between the independent variables using a correlation matrix. As expected, there were extremely high correlations—almost 1.0—between minute-based and charge-based variables, such as Total_Day_Minutes and Total_Day_Charge, and similarly for Total_Eve, Total_Night, and Total_Intl. These correlations were obvious, as charges are calculated directly from usage. Therefore, I decided to keep only the charge variables for further analysis, because they reflect the actual financial impact on the customer and are more likely to drive churn decisions.

	Account Length	Voice Mail Messages	Day Minutes	Day Calls	Day Charge	Evening Minutes	Evening Calls	Evening Charge	Night Minutes	Night Calls	Night Charge	International Minutes	International Calls	International Charge	Customer Service Calls
Account Length	1														
Voice Mail Mes:	-0.003285039	1													
Day Minutes	0.027294716	0.020187375	1												
Day Calls	0.049966197	-0.001714622	-0.0195065	1											
Day Charge	0.027293631	0.020190854	0.99999995	-0.019501031	1										
Evening Minute:	0.013731075	0.017134923	-0.0010671	0.004223405	-0.001060409	1									
Evening Calls	0.027972841	0.009124203	0.02397929	0.013117225	0.023977571	-0.011699751	1								
Evening Charge	0.013753808	0.017141094	-0.0010843	0.004215994	-0.001077523	0.99999977	-0.011684099	1							
Night Minutes	-0.01529957	0.004910758	0.02087691	0.001657302	0.020871488	-0.020370817	-0.001629504	-0.020374606	1						
Night Calls	-0.024372352	0.007471198	0.01790375	-0.021283335	0.017911035	0.006592421	-0.005736107	0.00658255	0.013092155	1					
Night Charge	-0.0152914	0.004877521	0.02083167	0.001643368	0.020826233	-0.020388216	-0.001606969	-0.020392012	0.999999198	0.013085577	1				
International M	-0.020481745	0.008756495	0.00291512	0.032224343	0.002916771	-0.016893623	0.044643571	-0.016939313	-0.001721787	-0.014620053	-0.001701377	1			
International Ca	0.008254763	0.009975834	0.02366566	0.001420175	0.02366243	-0.011394623	0.010217684	-0.011407957	-0.014868368	0.003784129	-0.014850146	0.025508199	1		
International Ch	-0.0205555	0.008835347	0.00298699	0.03226796	0.00298862	-0.017003731	0.044652186	-0.017006804	-0.001737888	-0.01459979	-0.001717505	0.999992709	0.025635528	1	
Customer Servic	-0.021252408	-0.006105574	-0.0218876	-0.015085772	-0.02189533	-0.02075405	0.007680288	-0.020768818	-0.041359074	-0.01613222	-0.041350228	0.004979737	-0.017527201	0.00489119	1

Part (c): Churn Relationship Analysis

To understand how various variables affected churn, I created PivotTables for each key variable and calculated the churn rate for each group. The churn rate was determined as the proportion of customers who left ("Yes" in the Churn column) within each category or bin. For example, churn increased significantly with the frequency of customer service calls. Customers with fewer than three calls had churn rates below 15%, while those with four or more calls showed churn rates over 40%, reaching as high as 100% in some small groups. This suggested that excessive contact with customer support could be a strong indicator of dissatisfaction.

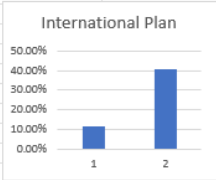
Count of Churn	Column Labels			Customer service calls	Churn Rate >30%	Overal Churn Rate	14.18%
Row Labels	No	Yes	Grand Total	Churn Rate			
0	377	55	432	12.73%			
1	665	79	744	10.62%			
2	430	55	485	11.34%			
3	239	24	263	9.13%			
4	59	43	102	42.16%			
5	17	27	44	61.36%			
6	8	8	16	50.00%			
7	1	3	4	75.00%			
8	1	1	2	50.00%			
9		2	2	100.00%			
Grand Total	1797	297	2094	14.18%			



A similar pattern was found in the international plan analysis. Customers who subscribed to the international plan had a churn rate of 40.6%, compared to only 11.4% for those who didn't. This suggested that the plan might not be meeting customer expectations.

On the other hand, having a voicemail plan was associated with significantly lower churn, just 8.1% compared to 16.4% for customers without it. Another increase was related to Total_Day_Charge: customers who had charges exceeding £50 had churn rates ranging from 35% to over 80%, while those with lower charges had much lower churn.

Count of Churn	Column Labels			International Plan
Row Labels	No	Yes	Grand Total	Churn Rate
No	1680	217	1897	11.44%
Yes	117	80	197	40.61%
Grand Total	1797	297	2094	14.18%

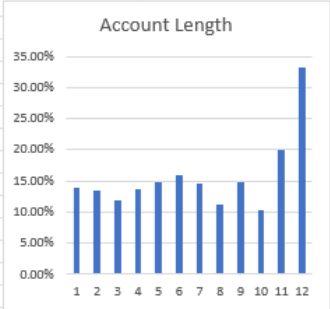


Count of Churn	Column Labels			voice mail plan
Row Labels	No	Yes	Grand Total	Churn Rate
No	1284	252	1536	16.41%
Yes	513	45	558	8.06%
Grand Total	1797	297	2094	14.18%

Part (d): Business Recommendations

Based on these observations, I made specific business recommendations. First, the company should contact customers who frequently call for service before they churn. These customers are likely experiencing ongoing issues and could be retained through improved support or incentives. Second, the international plan needs a review, possibly because its pricing or included services are not competitive. Conducting a customer survey could provide more clarity. Third, promoting voicemail service, even with free trials, could boost retention, especially for new or passive users. For frequent daytime users with high costs, I suggest using predictive bill notifications or offering capped-charge options. Finally, account length was not a strong predictor of churn on its own; newer customers (under 20 months) showed slightly higher churn. Implementing a loyalty reward programme around the 6- to 12-month period might help secure their long-term commitment.

Count of Churn	Column Labels			Account Length
Row Labels	No	Yes	Grand Total	Churn Rate
1-20	43	7	50	14.00%
21-40	96	15	111	13.51%
41-60	169	23	192	11.98%
61-80	258	41	299	13.71%
81-100	316	55	371	14.82%
101-120	336	64	400	16.00%
121-140	288	49	337	14.54%
141-160	159	20	179	11.17%
161-180	81	14	95	14.74%
181-200	35	4	39	10.26%
201-220	12	3	15	20.00%
221-240	4	2	6	33.33%
Grand Total	1797	297	2094	14.18%



In summary, this part of the analysis showed that churn is not random; it is driven by clear usage patterns and service features. By understanding and acting on these patterns, the company can reduce churn and build stronger customer relationships.

Problem 2: Regression Analysis and Forecasting

In this section, I have created and evaluated a regression model to understand what drives clothing sales for a fashion retailer and to forecast future performance. The dataset contained 23 years and included variables such as whether there was a holiday sale that year, whether there was an economic downturn, and a general time trend. The target variable was Clothes_Sold.

Part (a): Model Development and Variable Selection

To answer the given questions, I ran a multiple regression with Clothes_Sold as the dependent variable, and Holiday_Sale_Dummy, Recession_Dummy, and Trend_Index as independent variables. The first regression showed the following results:

- The coefficient for Holiday_Sale_Dummy was 358.30 with a p-value of 0.0057, indicating a strong and statistically significant positive effect.
- The Recession_Dummy had a coefficient of –90.95 but a p-value of 0.428. This makes it not significant in statistical terms.
- The time trend variable had a coefficient of 28.10 with a p-value of 0.0033, which also showed strong significance as the coefficient for Holiday_Sale_Dummy.

Since the recession dummy was not statistically significant at the 10% level, I excluded it from the final model. The model with Holiday_Sale_Dummy and Trend_Index had an R-squared value of 0.545, meaning it explained about 54.5% of the variance in clothing sales.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.738238458
R Square	0.544996021
Adjusted R Square	0.473153287
Standard Error	261.4788412
Observations	23

ANOVA

	df	SS	MS	F	gnificance F
Regression	3	1555982.975	518660.9915	7.58596	0.00156
Residual	19	1299052.504	68371.18441		
Total	22	2855035.478			

	Coefficients	Standard Error	t Stat	P-value	ower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	120.9776262	143.0468546	0.845720282	0.40823	-178.423	420.378	-178.423	420.378
Holiday Sales dumr	358.2952022	114.9372845	3.117310487	0.00567	117.729	598.862	117.729	598.862
Recession dummy	-90.94766073	112.3701753	-0.809357647	0.42833	-326.141	144.246	-326.141	144.246
Year	28.09953829	8.36522663	3.359088705	0.0033	10.5909	45.6082	10.5909	45.6082

Part (b): Model Interpretation and Diagnostics

The regression results showed that holding a holiday sale is one of the most effective ways to increase sales, contributing an average increase of 358 units. The trend over the years was also clearly positive, with sales increasing by 28 units annually. Although not statistically significant, the point of impact provided an initial point for prediction. The model appeared reliable, with no serious outliers (all standardised residuals were within ±2). This confirmed that the model was stable and reasonably good at predicting sales.

RESIDUAL OUTPUT

Observation	edicted Clothes soi	Residuals	Standard Residuals
1	58.1295038	40.8704962	0.168192993
2	444.5242443	-71.52424432	-0.294341342
3	114.3285804	-27.32858038	-0.112464397
4	233.3757794	179.6242206	0.739201577
5	528.8228592	-362.8228592	-1.493112838
6	289.574856	7.425144018	0.030556448
7	585.0219358	-235.0219358	-0.967177951
8	254.8262718	380.1737282	1.564516292
9	282.9258101	121.0741899	0.498252584
10	401.9730091	2.026990857	0.008341608
11	788.3677497	-36.36774967	-0.14966299
12	458.1720857	-384.1720857	-1.5809706
13	395.3239633	-20.32396328	-0.08363853
14	781.7187038	327.2812962	1.346849827
15	542.4707006	166.5292994	0.685312484
16	928.8654411	271.1345589	1.115791027
17	507.7221164	-277.7221164	-1.142900584
18	626.7693155	265.2306845	1.09149501
19	563.921193	-58.92119302	-0.242476425
20	1041.263594	-341.2635943	-1.404390713
21	978.4154718	448.5845282	1.846044981
22	739.1674686	-130.1674686	-0.535673852
23	676.3193462	-264.3193462	-1.087744609

Part (c): Probability Forecasting

If selling more than 1,000 units in Year 24, there would be a holiday sale and no downturns. Using the regression equation:

Predicted sales = 120.98 (intercept) + 358.30 (holiday) + 28.10 * 24 (trend)

= 1153.68 units

With a standard error of 261.48, the Z-score was calculated as:

Z = (1000 – 1153.68) / 261.48 = –0.588

Using the standard normal distribution, the probability of exceeding 1,000 units was:

= 1 – NORM.S.DIST(–0.588, TRUE) ≈ 72.2%

This high probability suggests strong expected sales.

Clothes sold	Holiday Sales	Economic Downturn/Recession	Holiday Sales dummy	Recession dummy	Year
99	No	Yes	0	1	1
373	Yes	Yes	1	1	2
87	No	Yes	0	1	3
413	No	No	0	0	4
166	Yes	Yes	1	1	5
297	No	No	0	0	6
350	Yes	Yes	1	1	7
635	No	Yes	0	1	8
404	No	Yes	0	1	9
404	No	No	0	0	10
752	Yes	No	1	0	11
74	No	No	0	0	12
375	No	Yes	0	1	13
1109	Yes	Yes	1	1	14
709	No	No	0	0	15
1200	Yes	No	1	0	16
230	No	Yes	0	1	17
892	No	No	0	0	18
505	No	Yes	0	1	19
700	Yes	No	1	0	20
1427	Yes	Yes	1	1	21
609	No	No	0	0	22
412	No	Yes	0	1	23
			1	0	24

Predicted Sales	1153.6617
Z Score	-0.5877
Probability	72%

Part (d): Business Insights and Recommendations

Based on the regression analysis and forecast, several business recommendations can be made. First, holiday sales should be prioritised as they have a significant impact on overall sales. Preparing early, running effective advertising, and managing stock during promotional periods can boost performance. Second, the positive trend over time indicates that the brand is growing. Continuing this trend will likely require investments in seasonal marketing, web presence, and brand awareness. Third, since economic downturns were not statistically significant, the company does not need to

drastically change its strategy during recessions but should monitor economic signals and consider temporary adjustments.

Lastly, with a predicted 72.2% chance of exceeding 1,000 units in Year 24, the business should plan for a high-demand season. Inventory, staffing, and marketing efforts should align with this expectation. To enhance future models, I recommend gathering more detailed monthly data and tracking additional factors such as pricing, competition, or regional trends.

Coursework Report: Problem 3 - Linear Programming and Sensitivity Analysis

I have used Excel Solver for the optimisation and performed additional sensitivity analysis to figure out how changes in requirements might affect the results.

Part (a): Mathematical Model Formulation

The main point is to minimise the total cost of getting components from three different suppliers. Each supplier charges a different price per unit and provides a specific percentage for all, small, medium, and large components in each unit supplied. The company has monthly requirements that must be met at least 500 large components, 300 medium components, and 300 small components. Additionally, the company cannot order more than 500 units from any one supplier due to availability limitations.

To build the mathematical model, I began by defining the decision variables:
Let X1 represent the number of units ordered from Supplier 1, X2 from Supplier 2, and X3 from Supplier 3. These are the values we need to find to minimise costs.

The objective function is:
Minimize $Z=20*X1+16*X2+12*X3$
This expression represents the total cost, where £20, £16, and £12 are the costs per unit from Supplier 1, 2, and 3, respectively.

Next, I set up the limits to reflect the company's component requirements. Each unit from a supplier provides a different percentage of each component type. Based on the given data:

- Supplier 1: 40% small, 40% medium, 20% large

- Supplier 2: 30% small, 35% medium, 35% large
- Supplier 3: 20% small, 20% medium, 60% large

Using these, the limits become:

- Large components: $0.20*X1+0.35*X2+0.60*X3 \geq 5000$
- Medium components: $0.40*X1+0.35*X2+0.20*X3 \geq 3000$
- Small components: $0.40*X1+0.30*X2+0.20*X3 \geq 3000$

In addition, we have the supplier limitations:

- $X1 \leq 500$
- $X2 \leq 500$
- $X3 \leq 500$

Lastly, the model must include the non-negativity limits:
$X1, X2, X3 \geq 0$

This entire formulation combines the real-world limitations and goals of the business. The next step is to implement and solve it using Excel Solver.

Part (b): Optimal Solution

To solve the model, I entered the objective function and limits into Excel and used Solver to find the optimal values of X1, X2, and X3. Solver was set to minimise the total cost. After solving, Solver returned the following optimal solution:

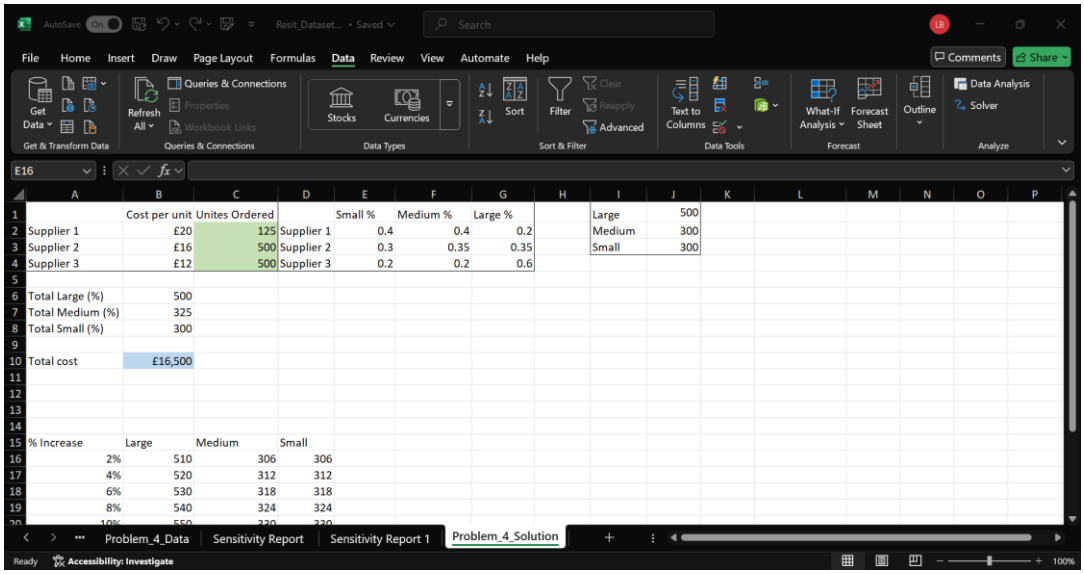
- 125 units should be ordered from Supplier 1
- 500 units from Supplier 2
- 500 units from Supplier 3

The total cost, calculated as 20(125)+16(500)+12(500)20(125) + 16(500) + 12(500), equals £16,500. The solution also satisfies the component limitations:

- Large components total: 0.20(125)+0.35(500)+0.60(500)=5000.20(125) + 0.35(500) + 0.60(500) = 500

- Medium components total: 0.40(125)+0.35(500)+0.20(500)=3250.40(125) + 0.35(500) + 0.20(500) = 325
- Small components total: 0.40(125)+0.30(500)+0.20(500)=3000.40(125) + 0.30(500) + 0.20(500) = 300

This solution minimises the total cost while satisfying all limitations. Notably, Suppliers 2 and 3 are used to full capacity, as they offer lower costs. Supplier 1 is used only partially, just enough to fulfil the remaining component requirement.



		Cost per unit	Unites Ordered		Small %	Medium %	Large %
Supplier 1		£20	125	Supplier 1	0.4	0.4	0.2
Supplier 2		£16	500	Supplier 2	0.3	0.35	0.35
Supplier 3		£12	500	Supplier 3	0.2	0.2	0.6
Total Large (%)		500					
Total Medium (%)		325					
Total Small (%)		300					
Total cost		£16,500					
% Increase		Large	Medium	Small			
	2%	510	306	306			
	4%	520	312	312			
	6%	530	318	318			
	8%	540	324	324			

Large	500
Medium	300
Small	300

Part (c): Model Revision for Sensitivity Analysis

The company wanted to study how increases in component requirements would affect the total cost and usefulness of the solution. I created an improved model that allowed the minimum required quantities of large, medium, and small components to increase simultaneously by the same percentage, ranging from 2% to 20% in increments of 2%.

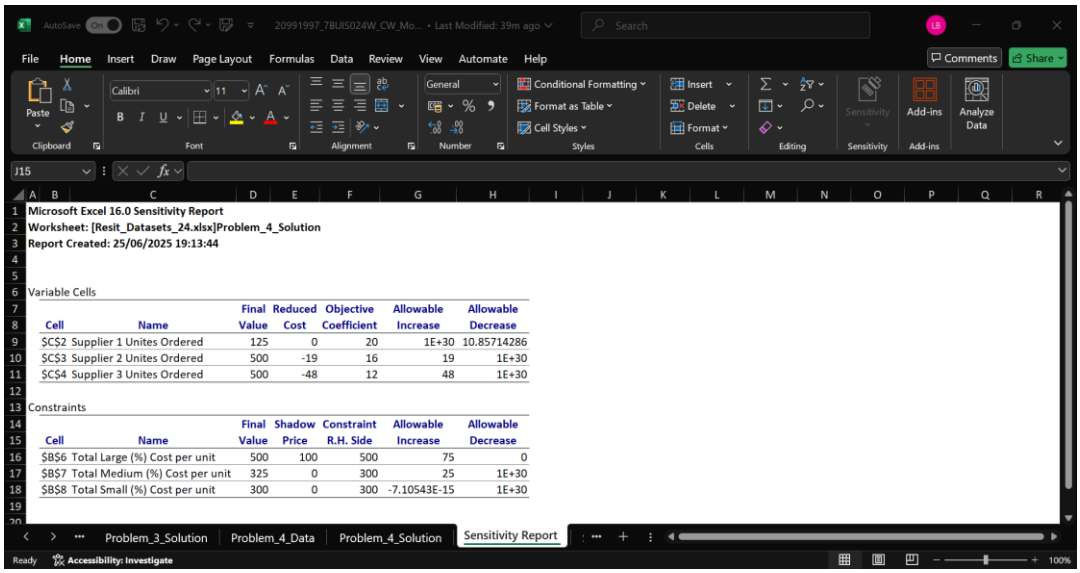
For each percentage increase, I updated the limited values accordingly and re-ran Solver. The table below summarises the findings:

% Increase	Large	Medium	Small	Total Cost	Feasible?
0%	500	300	300	£16,500	Yes
2%	510	306	306	£17,500	Yes
4%	520	312	312	£18,500	Yes
6%	530	318	318	£19,500	Yes
8%	540	324	324	£20,500	Yes
10%	550	330	330	£21,500	Yes
12%	560	336	336	£22,500	Yes
14%	570	342	342	£23,500	Yes
16%	580	348	348	–	No
18%	590	354	354	–	No
20%	600	360	360	–	No

Up to a 14% increase, the model remains feasible and provides optimal solutions. However, for 16%, 18%, and 20%, Solver was unable to find feasible solutions. This is because the maximum total number of components that can be ordered is 1,500 (500 from each supplier), and the higher needs are more than what the current components can provide within this limit. This shows a clear problem in the supply chain.

Part (d): Sensitivity Analysis Using Solver Reports

To further analyse the model, I used the Sensitivity Report generated by Solver.



Microsoft Excel 16.0 Sensitivity Report
Worksheet: [Resit_Datasets_24.xlsx]Problem_4_Solution
Report Created: 25/06/2025 19:13:44

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C2	Supplier 1 Unites Ordered	125	0	20	1E+30	10.85714286
C3	Supplier 2 Unites Ordered	500	-19	16	19	1E+30
C4	Supplier 3 Unites Ordered	500	-48	12	48	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
B6	Total Large (%) Cost per unit	500	100	500	75	0
B7	Total Medium (%) Cost per unit	325	0	300	25	1E+30
B8	Total Small (%) Cost per unit	300	0	300	-7.10543E-15	1E+30

(i) The shadow prices in the report indicate the minimal cost increase for each additional unit required. For large components, the shadow price was £100, and £0.00 for medium components and small components. Large components show the largest shadow price, meaning increasing the large component requirement by one unit raises the total cost to £100. Therefore, if the company were to

increase just one requirement that is set at 10 units, it must choose medium components for its option, for their shadow price is equal to zero and would not raise the cost for it.

(ii) The report identified all requirements, both for large components and small components, as binding constraints along with maximum capacity constraints for Suppliers 2 and 3. Because these constraints are precisely met in the best solution, they are tight. The overall price will change if the needs or skills rise. Also, the structure of the solution would be impacted. This means the company cannot order more from them without changing the structure of the solution or incurring higher costs.

(iii) The shadow prices remain valid only within a certain range. The requirement of large components can increase by up to 75 or decrease by 0. After that point, the shadow price of £100 is no longer valid. The allowable increase for small components is zero. Increases for those medium components do have an essentially infinite range. This indicates that they do not have an impact on the cost. The range of medium components is essentially infinite because they increase, which indicates they do not impact cost. These ranges are of help to the company in its planning. The entire solution does not need recalculating after modest changes within the range.

(iv) To estimate what is the cost impact of a 15% increase in all component requirements, I used shadow prices. A 15% increase in the large component needed increases from 500 to 575 units, which would add 75 units x £100= £7,500 to the total cost. The small component of the shadow prices is zero. The components for the medium add absolutely no cost. The shadow prices may no longer hold because this increase reaches the upper bound of the range, as the real cost could be higher still. The problem becomes infeasible after the 14% increase seen in part (c).

Through this problem, I have indeed formulated a linear programming model, solved it by using Excel Solver, and identified the best combination of supplier orders to minimise cost while meeting all component requirements and staying within supplier capacity limits. Following demand increases of 14%, the model reveals that the constraints are unmet as the current supplier structure fails. Furthermore, the sensitivity analysis does offer valuable insights into just how the cost is affected by small changes as well as which constraints happen to be tight. This shows how tools like linear programming and sensitivity analysis can support smarter, data-driven decisions in supply chain management.